PG No. 47 $T_{\mathrm{h}}(c)$ $m\bar{3}$ [cubic]

Table 1: Wyckoff site: `1o`, site symmetry: `m-3.`

No.	position	mapping
1	[0, 0, 0]	`[1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,21,22,23,24]`

Table 2: Wyckoff site: `6a`, site symmetry: `2mm.`

No.	position	mapping
1	$[x, 0, 0]$	`[1,4,14,15]`
2	$[-x, 0, 0]$	`[2,3,13,16]`
3	$[0, x, 0]$	`[5,8,18,19]`
4	$[0, -x, 0]$	`[6,7,17,20]`
5	$[0, 0, x]$	`[9,12,22,23]`
6	$[0, 0, -x]$	`[10,11,21,24]`

Table 3: Wyckoff site: `8b`, site symmetry: `.3.`

No.	position	mapping
1	$[x, x, x]$	`[1,5,9]`
2	$[-x, -x, x]$	`[2,7,12]`
3	$[-x, x, -x]$	`[3,8,10]`
4	$[x, -x, -x]$	`[4,6,11]`
5	$[-x, -x, -x]$	`[13,17,21]`
6	$[x, x, -x]$	`[14,19,24]`
7	$[x, -x, x]$	`[15,20,22]`
8	$[-x, x, x]$	`[16,18,23]`

Table 4: Wyckoff site: `12c`, site symmetry: `m..`

No.	position	mapping
1	$[0, y, z]$	`[1,16]`
2	$[0, -y, z]$	`[2,15]`
3	$[0, y, -z]$	`[3,14]`
4	$[0, -y, -z]$	`[4,13]`
5	$[z, 0, y]$	`[5,20]`
6	$[z, 0, -y]$	`[6,19]`
7	$[-z, 0, y]$	`[7,18]`
8	$[-z, 0, -y]$	`[8,17]`
9	$[y, z, 0]$	`[9,24]`
10	$[-y, z, 0]$	`[10,23]`
11	$[y, -z, 0]$	`[11,22]`
12	$[-y, -z, 0]$	`[12,21]`

Table 5: Wyckoff site: 24d, site symmetry: 1

No.	position	mapping
1	$[x, y, z]$	[1]
2	$[-x, -y, z]$	[2]
3	$[-x, y, -z]$	[3]
4	$[x, -y, -z]$	[4]
5	$[z, x, y]$	[5]
6	$[z, -x, -y]$	[6]
7	$[-z, -x, y]$	[7]
8	$[-z, x, -y]$	[8]
9	$[y, z, x]$	[9]
10	$[-y, z, -x]$	[10]
11	$[y, -z, -x]$	[11]
12	$[-y, -z, x]$	[12]
13	$[-x, -y, -z]$	[13]
14	$[x, y, -z]$	[14]
15	$[x, -y, z]$	[15]
16	$[-x, y, z]$	[16]
17	$[-z, -x, -y]$	[17]
18	$[-z, x, y]$	[18]
19	$[z, x, -y]$	[19]
20	$[z, -x, y]$	[20]
21	$[-y, -z, -x]$	[21]
22	$[y, -z, x]$	[22]
23	$[-y, z, x]$	[23]
24	$[y, z, -x]$	[24]